

OSCAR NDUGBU

Staff Backend & Platform Engineer — AI Infrastructure & Distributed Systems

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PROFESSIONAL SUMMARY

Backend and platform engineer with **5+ years** of hands-on experience designing distributed systems, event-driven APIs, and production observability, built on top of **12+ years** of federal-scale data engineering at Nigeria's Universal Basic Education Commission (UBEC) — architecting systems across all 36 states. Specializes in Node.js/TypeScript backend architecture, PostgreSQL multi-tenant systems (Row-Level Security), and full-stack observability (Prometheus, Grafana, OpenTelemetry). Designs for explicit failure modes, zero-downtime deployment, and deliberate recovery paths, and applies machine learning where it directly improves a system decision. Author of three open-source engineering tools; architecture advisor to early-stage fintech founders.

CORE TECHNICAL SKILLS

Languages & Runtime	TypeScript, Node.js, Python, SQL, Bash
Backend & APIs	Fastify, Express, FastAPI, GraphQL, REST, gRPC, BullMQ, WebSocket
Data & Cloud	PostgreSQL (Row-Level Security), Redis, Prisma, Docker, Kubernetes, AWS, GCP, GitHub Actions
Observability / SRE	Prometheus, Grafana, OpenTelemetry, ELK Stack, Sentry, Distributed Tracing
AI / ML Infrastructure	XGBoost, LightGBM, CatBoost, LLM APIs (GPT-4, Claude), Ollama / GGUF Routing, ZK-Proofs
Systems Design	Multi-Tenant RLS, Event-Driven Architecture, Checkpointed Recovery, Circuit Breaking, AES-256

PROFESSIONAL EXPERIENCE

Infrastructure Consultant — Backend & Reliability Jan 2024 – Present
Independent Consulting · Remote

- Built an AI-assisted incident diagnosis tool that converts live server logs into ranked root-cause hypotheses, cutting diagnosis time for client engineering teams while preserving a full evidence trail for review.
- Audited backend performance across fintech production systems, isolating connection-pool exhaustion, slow queries, and API latency spikes, then delivered metrics-driven remediation plans directly to CTOs and senior engineers.
- Designed proactive Prometheus/Grafana monitoring pipelines that surface anomalous behavior early, giving teams the lead time to intercept local faults before they escalate into outages.

Lead Backend Engineer — Encrypted Data Platform Jan – Jun 2024
Hashablanca · Remote

- Architected the core transfer backend for AES-256-encrypted payloads across four blockchain networks (Ethereum, Polygon, BSC, StarkNet), using idempotent retries and checkpointed state to guarantee zero data loss on interruption.
- Integrated Zero-Knowledge proof pipelines to validate sensitive data without exposing it, building tested boundaries between privacy-preserving reporting and the underlying payload.

Senior Data & Systems Engineer — Planning, Research & Statistics 2014 – Present (12+ yrs)
Universal Basic Education Commission (UBEC) · Federal Government of Nigeria · Abuja HQ / Lagos-based

- Built automated ETL pipelines that collect, validate, and consolidate education statistics from all 36 Nigerian state offices into a single audit-ready database, eliminating weeks of manual spreadsheet work every reporting cycle.
- Developed real-time analytics dashboards in Python and Power BI that give senior federal planners direct visibility into national education metrics, feeding budget-allocation decisions at the national level.
- Automated anomaly detection across state-level data submissions, tightening the validation pipeline while preserving human review for edge cases.

SELECTED PRODUCTS BUILT

TaxBridge — SME Tax Compliance & Filing Platform

Fastify 5 · Java 17 / Spring Boot 3 · PostgreSQL 15 (RLS) · Redis 7 · BullMQ · React Native / Expo SDK 54

- Architected a replay-safe tax filing workflow for Nigerian SMEs, enforcing tenant isolation at the database engine with PostgreSQL Row-Level Security instead of application-layer filtering.
- Designed idempotent BullMQ jobs keyed to submission content to absorb tax-authority API rate limits during deadline bursts, removing duplicate-filing risk on retry.
- Built a hash-chained audit trail for inspectable retry history and shipped a companion React Native/Expo mobile client; NRS DigiTax integration is in progress.

SabiScore — ML Prediction & Observability Platform

FastAPI · XGBoost / LightGBM / CatBoost · Redis 7 Pub/Sub · Prometheus & Grafana · Next.js 15

- Built an ensemble ML inference platform with versioned Redis caching and Pub/Sub fan-out, sustaining sub-50ms event delivery under concurrent production load.
- Designed a labeled fallback path that returns a lower-confidence prediction instead of stale output or a hard error when the cache or feature store fails.
- Instrumented Prometheus/Grafana telemetry across a rolling 90-day window to trace degraded responses back to their root cause.

SwarmXQ — Multi-Agent AI Orchestration Platform

Python · FastAPI · Ollama (GGUF) · Next.js 15 · pgvector

- Built a multi-agent orchestration system with triadic LLM routing that assigns each task to the right local model within an 8 GB memory budget.
- Designed checkpointed state and bounded fallbacks so interrupted agent runs resume from the last consistent checkpoint instead of restarting from zero.
- Open-sourced the underlying routing pattern as llm-dispatch, extending the approach beyond SwarmXQ itself.

The Yap Engine — AI Video Generation Pipeline

Python 3.11+ · Fastify 5 · TypeScript · Ollama

- Built a short-form video generation pipeline that stages work from creative brief through publish-ready output, routing each stage to task-specific local models.
- Enforced explicit queue, timeout, and resource-pressure boundaries so long-running generation jobs degrade deliberately instead of failing silently on constrained CPU/RAM hardware.

OPEN SOURCE & LEADERSHIP

- **pg-tenant** — Database-level tenant scoping built on PostgreSQL RLS, replacing application-level filters with explicit, testable isolation.
- **audit-chain** — Hash-linked audit sequence for fintech compliance; any later edit breaks the verification chain at the affected record.
- **node-debug-llm** — AI DevOps tool that converts live logs and traces into ranked root-cause hypotheses while preserving underlying evidence.
- Architecture advisor to early-stage fintech founders on system design, ML deployment, and security posture pre-launch.
- Speaker at Lagos JS and Python meetups on distributed systems and production ML; technical author at scardubu.dev.

CERTIFICATIONS & EDUCATION

AWS Certified Developer (2023) · **GCP Associate Cloud Engineer** (2023) · **OpenJS Node.js Services (JSNSD)** (2024) · **PostgreSQL 14 Associate** (2024)

B.Tech, Environmental Technology — Federal University of Technology, Owerri (2006–2011)